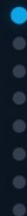


The Power of Vaccination

From Smallpox Eradication to COVID-19 in Record Time

1796 – 2020 • Source: Our World in Data



From 100 Years to Under 1 Year

Time from pathogen identification to vaccine licensure has collapsed — driven by mRNA technology, genome sequencing, and advanced microscopy.



100% Case Reduction Achieved for Diphtheria, Polio, and Smallpox in the US

Pre-vaccine vs. post-vaccine annual incidence and mortality (per million population). Data: Roush & Murphy, 2007.

DISEASE	PRE-VAX CASES	POST-VAX CASES	CASE REDUCTION	PRE-VAX DEATHS	DEATH REDUCTION
Diphtheria	158	0	100%	13.7	100%
Smallpox	250	0	100%	2.9	100%
Acute polio	141	0	100%	10	100%
Measles	3,044	0.2	99.99%	2.5	100%
Rubella	242	0.04	99.98%	0.09	100%
Pertussis	1,534	52	96.6%	30.8	99.7%
Hepatitis A	465	51	89%	0.5	88.7%
Acute hepatitis B	273	44	83.9%	1	83.6%
Pneumococcal	233	139	40.5%	24	31.3%

100% ELIMINATED

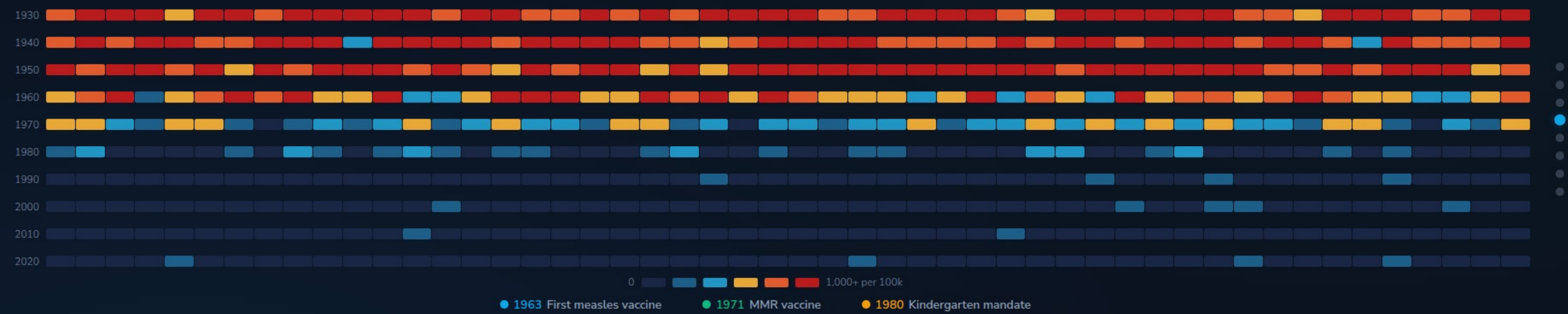
PARTIAL REDUCTION

SIGNIFICANT GAP



Measles Cases Collapsed Across All 50 US States After Vaccine Mandates

Simulated heatmap: state-level measles incidence per 100,000 population, 1929–2022.



Smallpox: The Only Human Disease Eradicated Through Vaccination

Certified eradicated in 1980 after a global campaign that stands as humanity’s greatest public health achievement.

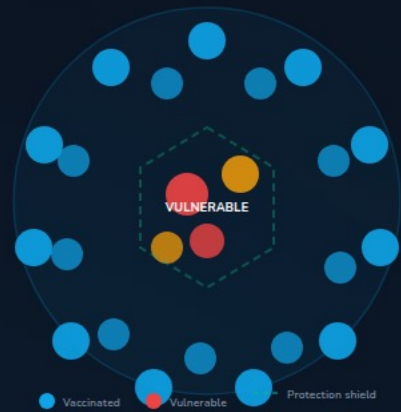
- 1959**
WHO passes resolution to eradicate smallpox worldwide
- 1967**
Intensified global eradication campaign launched
- 1977**
Last natural case occurs in Somalia
- 1980**
Smallpox declared eradicated worldwide

Why smallpox was uniquely eradicable

- 
Infected only humans — no animal reservoir
- 
Symptoms easily distinguishable
- 
Vaccine was cheap and effective

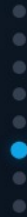
Herd Immunity: How Vaccinating Many Protects the Most Vulnerable

When enough people are vaccinated, pathogens cannot spread effectively — shielding those who cannot be vaccinated.



How herd immunity works

- Vaccinated individuals create a protective barrier that blocks pathogen transmission chains.
- Those who **cannot be vaccinated** — newborns, immunocompromised individuals, and cancer patients on chemotherapy — rely on community protection.
- When vaccination rates drop below the herd immunity threshold, outbreaks can spread to vulnerable populations.
- The threshold varies by disease: highly contagious diseases like measles require ~95% coverage, while others need less.



Gaps Persist: Pneumococcal Disease Shows Only 40% Case Reduction

Not all vaccine-preventable diseases are under control. These diseases need improved vaccines, better access, and higher coverage.



- Key Barriers
-  **Access gaps** — globally, not everyone has access to vaccines, especially in low-income countries.
 -  **Supply shortages** — production and distribution challenges put children at risk.
 -  **Trust deficits** — poor scientific understanding and misinformation reduce uptake.



Key Takeaways: Vaccines Have Eliminated Deadly Diseases — But the Job Isn't Done

- 01 Vaccines achieved 100% elimination of cases and deaths for diphtheria, smallpox, and both forms of polio in the US.
- 02 Development timelines have collapsed from over 100 years (typhoid) to under 1 year (COVID-19), driven by mRNA technology and genomic tools.
- 03 Measles cases dropped to near zero across all 50 US states following the 1980 kindergarten vaccination mandate.
- 04 Smallpox remains the only human disease eradicated through vaccination, certified in 1980 after a global campaign.
- 05 Coverage gaps persist for pneumococcal disease (40.5% case reduction), hepatitis A (89%), and hepatitis B (83.9%), highlighting the need for improved access, coverage, and newer vaccines.

Improve access, coverage, and trust in vaccination worldwide

The next chapter of vaccination success depends on global health systems reaching every community.